

Snatch and Hatch: Improving Receptivity Towards a Nature of Science with a Playful Mobile Application

Hannah Qiao
National University of Singapore
Singapore
hannah@ahlab.org

Hussel Suriyaarachchi
National University of Singapore
Singapore
hussel@ahlab.org

Sankha Cooray
National University of Singapore
Singapore
sankha@ahlab.org

Suranga Nanayakkara
National University of Singapore
Singapore
suranga@ahlab.org

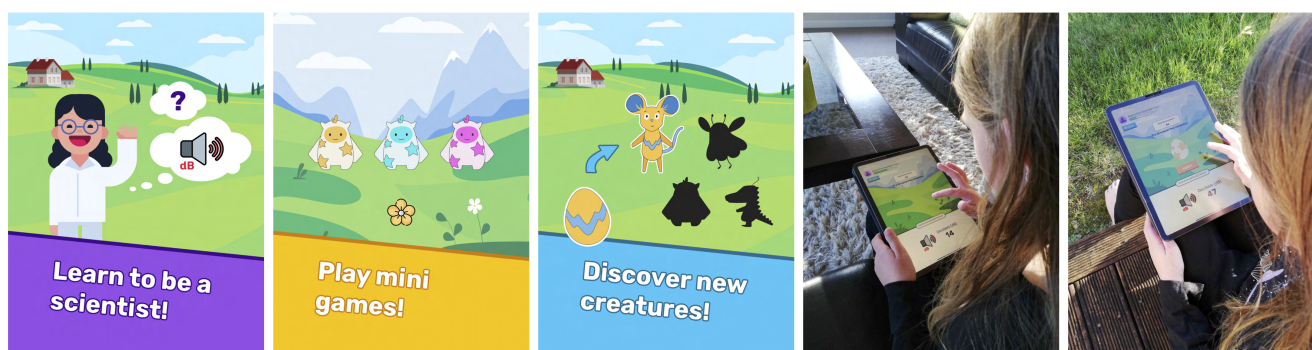


Figure 1: *Snatch and Hatch* provides a playful experience of capturing virtual creatures by manipulating the sound amplitude measured using the built-in microphone. We implemented a production-level tablet application and conducted a multi-day evaluation with children in a naturalistic setting.

ABSTRACT

Science literacy is an increasingly important skill in the 21st century. With engagement and motivation as vital precursors to learning science, we believe introducing children to playful interactions with scientific phenomena would improve their motivations and attitudes towards science. To investigate this, we developed a tablet application where children journey in a story-driven game to capture virtual creatures by manipulating the sound measured using the built-in microphone. This game was designed with feedback from 16 children and 10 parents. In this paper, we describe the iterative design process and findings in a multi-day study with 11 more children aged between 8 and 12. Children were motivated by the game, demonstrated a strong association between sound and its behaviour in the physical world, and expressed enthusiasm to learn more in the classroom.

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CCS CONCEPTS

• Applied computing → Interactive learning environments.

KEYWORDS

Nature of Science, Intrinsic Motivation, Children, Science Education, Mobile Game, Technology Enhanced Learning

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1 INTRODUCTION

From an early age, science is taught to children to spark competencies in problem-solving and critical thinking processes [21, 39]. An influential framework used by numerous educators to structure the instruction of scientific knowledge is the Nature of Science (NoS) [24], which introduces science as a discipline that is empirical, tentative, creative, social and subjective [21].

A particularly critical ingredient to instilling the tenets of NoS is motivation, as a motivated student demonstrates greater persistence and engagement in academic tasks and achieves better long-term outcomes [8, 11, 27]. Intrinsic motivation, in particular, can radically influence children's receptivity to instruction across various

learning contexts [3, 23, 35]. Inspired by constructivist views, researchers and practitioners have looked to promoting motivation in science learning through playful and game-based experiences. Examples include gamified virtual environments for investigating and solving scientific inquiry-driven problems [18] and virtual laboratories for performing experiments [17]. In the creation of such gamified experiences, it is crucial to balance any fun features with educational content.

Our overarching goals are twofold: to develop a playful experience to motivate children to the Nature of Science outside the classroom and to understand how this playful experience impacts their intrinsic motivations and attitudes toward science. In this paper, we describe our design, development and evaluation of *Snatch and Hatch*, a tablet application for delivering an engaging gaming experience entangled with concepts relevant to NoS. *Snatch and Hatch* was informed by a formative investigation with 16 children and 10 parents. In our current implementation, the *Snatch and Hatch* application leverages the internal microphone of smart devices as a sound amplitude sensor to provide a story-driven gaming experience where children can draw connections to their natural environment in real-time. We envisioned that this would improve children's interest in NoS by making the capturing and use of sensor data fun, engaging them more deeply with the sensors and their measurements. We implemented a robust, production-level tablet application with adequate scaffolding enabling *Snatch and Hatch* to be used by children in their own homes, without further requirements for setting up or supervision. We report the results of a multi-day study involving another 11 children. In summary, our contributions include:

- A formative investigation with children and parents which indicated the importance of hands-on exploration with playful experiences to achieve sustained engagement. For parents, it is important to communicate the educational value and safety considerations of using this digital application.
- Design & development of *Snatch and Hatch*, a story-driven game to capture virtual creatures by manipulating the sound amplitude measured using the built-in microphone. We implemented a production-level tablet application capable of reliably collecting rich data in a naturalistic setting.
- A multi-day user study with *Snatch and Hatch*. Our quantitative data shows that children displayed sustained engagement and intrinsic motivation. Observations and semi-structured interviews with children indicated they easily navigated the game and were keen to learn more about sound. Based on these insights, we propose that a carefully designed playful experience could condition children to be excited to learn when concepts are introduced more formally.

2 RELATED WORK

2.1 Sensor-Based Science Learning Platforms

Sensor-based learning technologies have gained significant popularity as they allow students to interact with real-world physical phenomena. Examples include web-based learning platforms with ready-to-use lessons [2, 4], plug-and-play sensors [5, 12, 20, 33, 34, 36] and custom-designed applications for data collection during field trips [32]. These allow students to develop scientific reasoning,

formulate hypotheses, conduct experiments, collect and analyse data, and reflect on their observations and findings at their own pace.

Herodotou et al. [16] describe their design and evaluation of a mobile app that allows students to run their own scientific investigations through acquiring and visualising data from all available internal sensors of the phone. Although students were interested in using the app and enjoyed the ability to use phone sensors to define their own scientific investigations, the authors were cautious in their claims regarding the app's ability to sustain students' interest and motivate long-term engagement. Similarly, Vogel et al. [40] shared reservations on their deployment of a prototype that combined a smartphone application with an external probing device. They specifically mention challenges relating to maintaining the infrastructure of their solution should it be deployed for real-world educational use. While these examples show potential in an application that enables students to engage in their own scientific investigations, motivating engagement and usage outside the classroom remain challenging. Further, designing an application that can support such engagement will likely require features beyond simple data collection and visualisation.

2.2 Games to Support Learning & Engagement

Educational games are a successful tool to engage and motivate students [13, 19, 22, 28]. Unlike most traditional teacher-led instruction, games can include various challenges, rewards, cooperation, competition, or a fantasy narrative, packaged into a single experience, to make learning more fun [22, 28]. In the field of science, examples of these games include those that target a specific subject area, such as learning organic chemistry through gameplay mechanics that mimic chemical reactions [14] or 'citizen science' games, which require crowd-sourced data to generate scientific knowledge on a variety of subjects [25, 26, 38].

Games can provide opportunities to build upon implicit learning, which describes students' understandings based on their everyday experiences [31]. While students' implicit knowledge may create misconceptions that do not match what is taught in schools, it creates opportunities for educators to teach explicit science knowledge [31]. For example, consider a common misconception among younger students, where 'light things float and heavy things sink'. While this may appear to match their personal experiences, it provides opportunities for teachers to teach about an object's buoyancy, density and effects on surface tension. Building upon this, our goal is to develop a playful experience with implicit learning [37] to support the development of explicit knowledge [7, 31]. In *Snatch and Hatch*, the playful interactions are mapped into characteristics of Nature of Science (NoS) that will be primarily used at home to supplement any learning in formal educational settings.

3 FORMATIVE INVESTIGATION

We conducted formative studies through interviews to understand how children spend their free time on digital devices, particularly in educational games and the area of science, and discover insights to inform our system design. In addition, we explore the perceptions held by parents toward educational games. Ethical approval for this

study was obtained from the Human Participants Ethics Committee of the National University of Singapore.

3.1 Method

We interviewed 16 children (aged between 8 and 14 years) and 10 parents. Multiple children from the same household were interviewed, however no two parents from the same household were interviewed. All participants were interviewed alone, regardless of whether they were from the same household or not. These interviews followed a semi-structured format and were recorded over a video call due to Covid-19 restrictions. The interviews with children involved querying their device usage, attitudes and participation in science activities in and out of the classroom. We noted if they possessed prior experience with educational apps or games and asked them to report the characteristics of a game they would find exciting. Parents were asked on their involvement in providing ways to occupy their children's free time outside of school, including if or how they manage screen time and their views on this. We also asked about their reasoning behind providing or allowing a digital activity.

Following the completion of these interviews, responses were transcribed by the research team. These responses were then qualitatively coded into common themes and grouped to identify patterns in responses across participants. Through this process, we identified three high-level themes that have direct implications on our subsequent design, as discussed below.

3.2 Key insights

Creating opportunities for hands-on explorations: Most children described their understanding of science as creating “rockets”, “volcanoes”, or other practical experiments they had previously completed at school. Although a few expressive responses emerged when students were prompted further, many remained unable to articulate understanding beyond their initial explanation. Inside classrooms, teachers are constrained by the content they need to cover as outlined by their curriculum. Unless the child is able to create their own unsupervised experiment, or be prompted to engage in science-based activities, they are unlikely to engage in such endeavours on their own. This observation was also reflected outside of school hours, where children tended to not engage in science-related activities, although this lack of engagement is typically compounded by the need for parental involvement and supervision. In response to these findings, we plan to 1) design a platform that can be used outside of formal educational settings, 2) requires little parental supervision, and 3) be accessible with familiar technology.

Playful experience and sustained engagement: While our interviews revealed little science learning, children appeared to enjoy and more readily recall the previously stated practical experiments compared to more theory-oriented classes. These practical experiments typically incorporated game-like aspects, such as competition between classmates, and visual and kinetically-based challenges. Such responses further supports existing successes in engaging children via these learning methods. Furthermore, playing games (offline and online) were a popular way for children to spend their free time

outside school. While some of these games (e.g. Roblox¹, Among Us², and Brawlstars³) were designed for entertainment rather than education, they are hugely popular among many children. When asked about these games, children expressed enjoyment of a multiplayer mode, customisation, variety in gameplay, regular new content, and levelling up characters or completing missions. These elements all reflect prior work that synthesises successful features in educational games [22, 28]. By complementing these insights with the key principles of the Nature of Science, we aim to devise a playful gamified experience to facilitate more frequent and long-term engagement.

Clear communication of educational value and safety: When parents were asked about educational apps, their attitudes were generally quite positive. However, educational value, online safety, and privacy lie among their primary concerns. Consequently, designing a platform for children to use at home requires explicitly communicating resolutions to these concerns with their parents. Since mobile tablets such as iPads are popularly perceived by parents and instructors as educative tools suitable for children, our design and development of a playful science learning application targeted this genre of devices.

4 SNATCH AND HATCH DESIGN

Snatch and Hatch is built upon a fictional world where children must help a scientist named ‘Professor Yuni’, who is researching new creatures in this world. These creatures, found as eggs, only appear and hatch by manipulating real-world phenomena recorded through smart devices. In this version, *Snatch and Hatch* uses the device microphone, which captures the amplitude of sound. Therefore, children must create varying levels of sound in the physical world in order to catch every creature. The sound amplitude range in which each creature can be captured is conveyed to children as ‘clues’ to encourage a hypothesis-driven discovery of appropriate sounds in their immediate environment. New creatures were released in regular intervals. As children progressed through the game, they would eventually become a ‘legendary scientist’ by gaining user experience points whenever they successfully engage in different features.

4.1 Iterative Design Process

The process of designing *Snatch and Hatch* involved several iterations of low-fidelity and high-fidelity prototyping through usability sessions. Of the 16 children originally interviewed, we observed 7 children (aged between 8 and 12 years) playing the game. The remaining children who were interviewed did not participate in this process due to being unable to participate in sessions in person. All usability sessions were completed in person and observations were recorded qualitatively by the researcher. Each usability session involved providing children with the same tasks to complete by using the same set of interfaces. For example, children were asked to ‘complete the tutorial’, ‘capture a second creature’ or ‘find out how many creatures there are to catch’. This process allowed us to observe any potential usability issues and reactions the children had

¹<https://www.roblox.com/>

²<https://www.innersloth.com/games/among-us/>

³<https://supercell.com/en/games/brawlstars/>

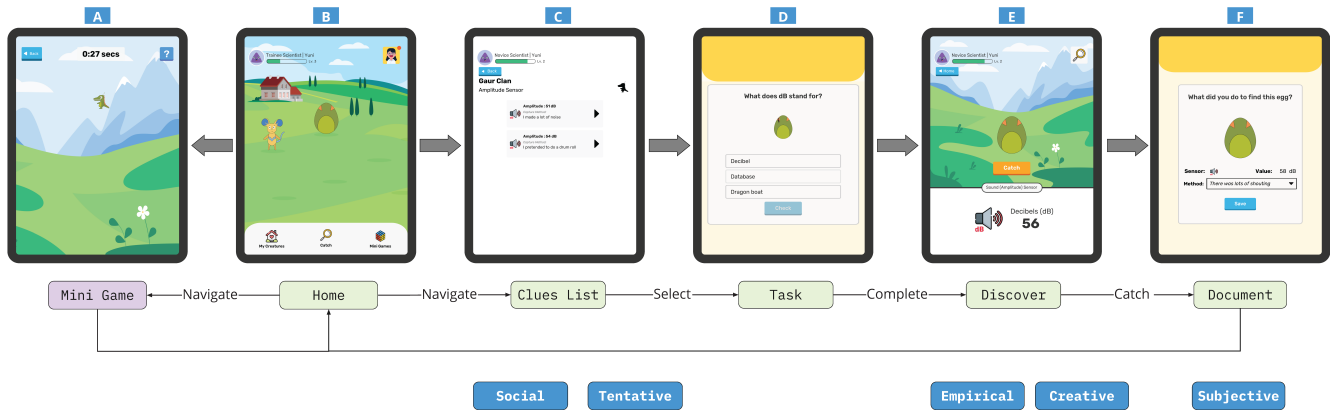


Figure 2: *Snatch and Hatch* User Flow, showing the key features of our application that were informed by tenets of Nature of Science.

as they played, such as whether they appeared to enjoy the game and be engaged in the features, or any other positive or negative reactions. We did not provide any external feedback throughout these sessions to see how children might navigate and play the game on their own.

Many insights, such as navigational enhancements, the importance of a carefully designed tutorial, or various aesthetic improvements were identified throughout this process. However, a key insight revealed how non-functional low-fidelity prototypes appeared to hinder some children’s responses. Many children commented on how “slow” or “broken” the game was, particularly when the sensor did not produce changing values in a static prototype. This caused confusion among most children, although this appeared to affect the younger children more than the older children. As a result, we also compared children’s feedback in the low-fidelity stages with additional usability feedback from adults, including an external industry expert who has worked with well-regarded children’s entertainment brands.

This process reinforced some of the key objectives needed to develop a children’s game, including: (1) having a clear and easy-to-understand goal; (2) the need to seamlessly integrate tenets of NoS into game play mechanics; (3) incorporating elements of fun for children to engage and be motivated to come back; and (4) the output (story narrative, creature designs, graphics, sounds, and animation) should feel like a game. For *Snatch and Hatch*, this process also helped us choose to focus on the sound sensor as a starting sensor. Sound was a recognisable concept to children and can be used to create playful experiences as shown in prior work [30].

4.2 Walk-Through of Key Features

Creature Clues: Clues to capture creatures are released in regular intervals. These clues are framed in a way that mimics another user’s findings. For example, the clues to capturing a particular creature whose sound amplitude range lies between 50-60dB include “I made a lot of noise” and “I pretended to do a drum roll”, representing actions that would likely cause loud sounds (Figure 2 C). This reflects the social tenet of NoS as children are, in essence, building upon preceding knowledge. This feature also serves to

reflect the tentative nature of science, as the decibel and method at which children might find a creature may not exactly match those provided in the clues. As children do not see the pre-defined decibel ranges of each creature, children can use clues and their own observations to build evidence for the range at which the creature might be found.

Creature Discovery: New creatures in *Snatch and Hatch* are discovered in the form of an egg (Figure 2 E). Through using the device microphone as the sound amplitude sensor, children must reach the correct decibel range to reveal the egg. They must come up with ideas to change the level of sound around them and try different methods if one does not work. This supports the tenet of NoS where science is creative and empirical, as children are creating their own observations of sound levels and need to be creative in order to generate different decibel levels. Additionally, it establishes that there is no one way to do science by allowing the required decibel level to be generated in multiple ways.

Creature Documentation: Once a creature egg is discovered, children must document their method of discovery. A dropdown allows children to either select a provided method or write their own (Figure 2 F). When documenting a method of discovery, they need to describe what they did or what had happened around them for the creature to appear. This feature promotes the useful skill of accurate record-keeping for reproducibility.

Creature Hatching: Each creature is first discovered as an egg and must be hatched to reveal the fully formed creature. Children are asked a question about sound amplitude at the beginning of every hatching stage. Examples of these questions include recalling a previous discovery method (Figure 2 D), comparing objects which might make louder or quieter sounds, or relating environmental conditions with the correct decibel value. These questions encourage children to recall prior knowledge and apply them in new scenarios. After answering the question correctly, the game then directs children to re-discover the egg (Figure 2 E) and re-document (Figure 2 F). These steps must be repeated anywhere between two to four times to fully hatch each egg. In doing so, children are continuously engaging with the creative, empirical and tentative tenets of NoS.

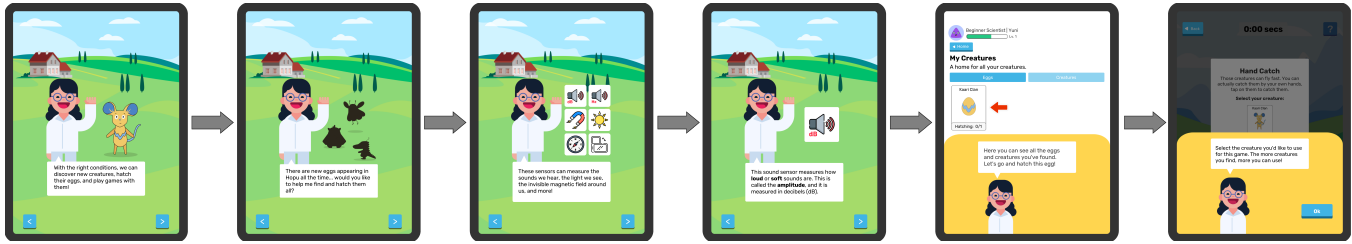


Figure 3: *Snatch and Hatch* Tutorial, which is integrated into the gameplay and alleviates the need to explicitly explain all the features.

Mini Games with Creatures: Three mini games were added to this version of *Snatch and Hatch*. These functioned primarily for children to have fun, gain user experience points, and sustain engagement between creature releases. However, only fully hatched creatures could be used in these games, which provided an incentive for children to find and hatch these creatures. The three mini games includes (1) tapping the creature moving across the screen (Figure 2 A), (2) revealing a creature moving in the dark with a torch, and (3) shaking the device to change a flower’s colour, to match the colour of the creature.

Introductory Tutorial: The tutorial for *Snatch and Hatch* was designed to be interactive and seamlessly integrate into gameplay, without the need to explain all the features from the beginning. Hence, features are introduced throughout the tutorial one at a time, rather than all at once. The tutorial introduces ‘Professor Yuni’ (Figure 3), who guides the user throughout the game and sets the scene for an overarching narrative. Animated arrows were also added as a visual signifier to direct children throughout the tutorial flow. By the end of the tutorial, children would have engaged in the core features of the game (i.e. creature discovery, documentation, and hatching) and be able to access any unlocked mini games, effectively giving them freedom to play on their own from thereon.

5 SNATCH AND HATCH IMPLEMENTATION

This section outlines our technical specifications and considerations for developing a software application for *Snatch and Hatch*, which can be handed over to children to play on their own. As such, the application needed to be robust and easy to use by children as well being able to collect reliable data without compromising the privacy of children. We adopted a modular development paradigm comprising a Unity⁴ game engine and an Amazon Web Services (AWS)⁵ based cloud infrastructure to drive the functionality. The Unity engine is responsible for defining the behaviour of *Snatch and Hatch*, including in-game physics, hardware control and logic operations. The overall system architecture is illustrated in Figure 4.

5.1 Game Play Front-end

User Interface: In implementing the user interface, we adopted a rigorous development and testing policy to ensure a fluid user experience with minimal room for error. Visual elements generated in the design phase were imported into a 2D gamespace and rendered

as Scenes using the Unity Editor. We intentionally maintained a 2D perspective, as introducing a third dimension may potentially overwhelm children during gameplay, thus shifting their focus and elevating the entry barrier. All game scenes were made responsive using native alignment grids, object layouts, and pivoting methods to support compatibility across various screen sizes.

Audio and Haptic Feedback: Non-visual feedback modalities, including haptics and audio, are now a standard across smart devices such as tablets. We looked to leverage these outlets as a means of presenting a more stimulating and immersive in-game experience. To oversee audio mixing, we implemented an Audio Service responsible for cueing background music and event-based sound effects. Alerts for specific events, for example, when a creature or user level-ups, were programmed as acute, high-fidelity vibrotactile responses on Unity’s haptic engine.

Device Hardware Usage: The fundamental gameplay of *Snatch and Hatch* revolves around the sound amplitude present in the microphone data of a student’s tablet. However, the software development kits of Unity, Android and iOS fail to natively provide this data in their microphone libraries. Extracting the amplitude values of the live microphone feed involved developing a time-window-based Fast Fourier Transform (FFT) algorithm to detect peaks in the audio spectrum. The subsequent amplitude data will then be available through a “virtual sensor” during the game. In addition to the microphone, our application relies on the tablet’s accelerometer sensor to detect a “shake”, which acts as the primary control in specific mini-games.

Game Control Logic: With most aspects of the game structured to lead to a goal state, we implemented several control logic components to manage user-centric tasks and their outcomes. For instance, amplitude-based creature discovery, task-based creature hatching and mini-game behaviour are all constructed upon boolean evaluators and primitive variable manipulation. Unity schedulers and co-routines were essential for translating the results of such operations into observable output in a computationally optimal and safe manner. The live display of amplitude readings and the creature release timer are among the features that rely on these processes for execution.

5.2 Content Delivery Back-end

To facilitate the opportunity of expanding the game as it progresses, we designed our system to serve all core game functions independently of their accompanying data. The absence of game content,

⁴<https://unity.com/>

⁵<https://aws.amazon.com/>

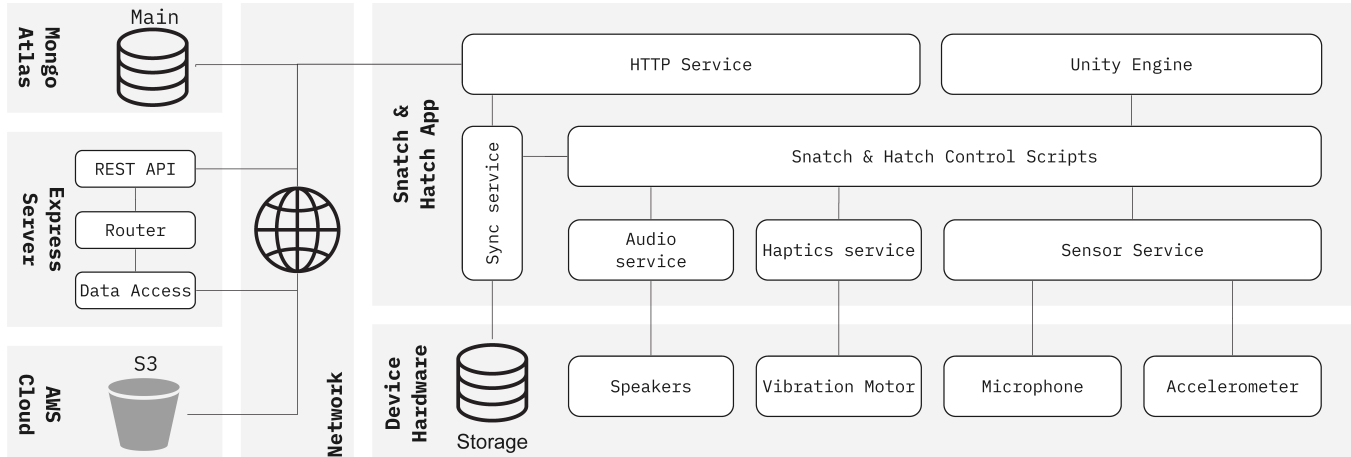


Figure 4: System Architecture of *Snatch and Hatch*. This robust framework enabled us to conduct a multi-day user study with children in a naturalistic setting.

particularly multimedia, makes for a relatively lightweight application bundle accessible via digital distribution platforms such as the Apple App Store and Google Play Store. This configuration allows for a pragmatic approach to issuing content rollouts by sparing the need for excessive app releases, which may hinder usability.

Database and Web Service: We organised all descriptive data containing the definitions of creatures, tasks and clues as JSON-styled objects in a NoSQL database hosted using the managed, multi-cloud database service offered by MongoDB Atlas⁶. Establishing communications between the database and the tablet application entailed implementing an Application Programming Interface (API) to effectively transmit any required data on-demand over the internet. An Express⁷ server built on the Node.js⁸ environment served as our Web API framework of choice to coordinate the transactions among the two entities. We packaged our server-side application into a Docker⁹ container and deployed it through the Elastic Container Service (ECS)¹⁰ available on AWS.

Media Storage Service: With *Snatch and Hatch* offering a gaming experience rich in 2D graphics, textures and sounds, using such assets demands an external media source to regulate the storage requirements of our tablet application. Consequently, we opted to use the S3¹¹ cloud storage solution of AWS as the media provider for our game. The metadata from applicable API responses, together with a secure service key, can then be used to dynamically construct the S3 identifier needed to fetch the desired media resource.

Data Sync Service and Offline Use: Considering the heavy dependency of our application on remote data sources, we incorporated a data sync service to support the stable management of in-game content. On initial application launch, the sync service enforces internet connectivity to clone the current state of the database and media storage onto the devices' persistent storage. We employ

Unity's standard mechanism of storing game data as key-value pairs in our application. JSON data retrieved from the database are held as serialised strings, while media files are encoded into their Base64 counterparts, ready to be parsed as textures, images or sounds. By utilising this strategy, we guarantee a robust system capable of reliably operating both online and offline.

With new and modified content appearing over time, a version control system tracks changes between the local and remote storage states. Our sync service relies on Unity's Remote Config¹² package to detect a change in the data version and trigger an update when adequate network access is detected. Thus, users can experience a seamless evolution in their game world with no additional intervention.

6 SNATCH AND HATCH EVALUATION

Our evaluation aimed to investigate the usability of *Snatch and Hatch* and assess its efficacy in motivating children and creating a positive attitude towards learning science.

6.1 Participants & Procedure

We recruited 11 children (5 boys, 6 girls) aged between 8 to 12 years ($M = 9.8, SD = 1.2$) from our network through the university. Consent for the children to participate were provided by their respective parents and ethical approval for this study was obtained from the Human Participants Ethics Committee of the National University of Singapore. None of these participants were involved in the iterative design phase of *Snatch and Hatch*. All children were enrolled in a science curriculum at school and conveyed a trivial understanding when enquired about the concept of sound. Although some children reported having prior experience in educational games, this was their first time interacting with live sound data from their environment in a game-based setting. All participants owned an iPad that could be used to run *Snatch and Hatch* at home. The study consisted of three parts:

1. Introduction (30 mins): The study commenced with a brief face-to-face interview where we asked questions about the usage of

⁶<https://www.mongodb.com/atlas>

⁷<https://expressjs.com/>

⁸<https://nodejs.org/en/>

⁹<https://www.docker.com/>

¹⁰<https://aws.amazon.com/ecs/>

¹¹<https://aws.amazon.com/s3>

¹²<https://unity.com/products/remote-config>

digital devices and general interest in science and technology. We then directed them to the Apple App Store, where they could install *Snatch and Hatch*. With no further guidance, we asked the child to begin playing and observed their initial responses and interactions with *Snatch and Hatch*.

2. Free Gameplay (4 days): Children were told that they can continue to play *Snatch and Hatch* as they wished over the next four days. We made it evident that children had complete freedom in deciding whether or not they would return to the game and the extent to which they would play it. Both the child and their parents were informed of our data collection mechanisms, such as anonymous usage tracking and in-game surveys, to which they consented. Participants were asked to complete a short questionnaire on intrinsic motivation (which pops-up in the game itself) twice during this period.

3. Conclusion (30 mins): At the end of the fourth day, we conducted a follow-up interview to understand children's experience with the *Snatch and Hatch* game.

6.2 Metrics and Collected Data

Usability. To discover usage problems and opportunities to improve our design, we conducted usability tests. We needed to use an appropriate method to collect usability data from children as their cognitive capabilities are different from adults. Hanna et al. [15] suggested that think-a-loud may not work for children under 12 years and as such, we collected usability data from our observations during introductory session and post-task interviews which has been shown to be as effective as think-a-loud [10].

Intrinsic Motivation & Attitude Towards Science. To explore the influence of our application on children's motivation, we administered a Likert scale questionnaire using the interest/enjoyment and value/usefulness subscales of the Intrinsic Motivation Inventory (IMI) [1]. IMI is a well-evidenced survey instrument for evaluating learner motivation, with the interest/enjoyment sub-scale considered the self-report measure of intrinsic motivation, while the value/usefulness sub-scale serves as an indicator of self-regulation towards a learning activity. We distributed two identical questionnaires, the former once children collect their second creature and the latter after capturing the fifth creature. Children would be prompted to complete these questionnaires (see Appendix for the questionnaire) within the *Snatch and Hatch* application, which features an integrated Qualtrics¹³ workflow to enable the secure management of survey responses. In addition, our semi-structured interviews provided qualitative finding to supplement the IMI.

In-Game Interactions and Analytics. Children's interactions with the *Snatch and Hatch* application offer a rich avenue to understanding their learning experiences and perceptions as they unfold. Using the content delivery system of *Snatch and Hatch* described in Section 5.2, we integrated an automated data collection mechanism capable of maintaining a cohesive record of user interactions as children progress in their gameplay. Our web service workflow enables the transmission of relevant data from the tablet application to the cloud database when specific event triggers are detected in the

game. Navigation button clicks, active screen time and data entry events serve as the markers in determining usage and inferring children's experiences. With data privacy and user safety a priority, we substituted all sensitive, user-specific information with an autogenerated identifier prior to any data storage or transfer, thus preserving anonymity and safeguarding the user.

7 RESULTS & DISCUSSION

This deployment of *Snatch and Hatch* depicts its first-ever use with a child audience in a real-world context. Our evaluation was commissioned to address the objectives stated in Section 1, specifically, the usability of our application, its effect on intrinsic motivation and on children's attitudes towards science. To assess usability, we reflect on our observations of children participating in gameplay and their feedback concerning the perceived ease of use. We gauge motivation through quantitative data obtained from responses to the interest/enjoyment subscale of the IMI and application usage logs, along with qualitative insights from interviews with children. Finally, for attitudes towards science learning, we analyse the questionnaire data on the value/usefulness subscale of the IMI and the follow-up interviews conducted at the end of the study.

All 11 children completed the first part of the study and participated in follow-up interviews. During our post-interviews, we found that only eight children continued to play during the four day free-play period at home. Three children did not continue to play after the initial introduction. P9 (12 years old), who did not play at all, said "*the art and features are for younger kids*" and "*it was too simple*". The 8 children who used *Snatch and Hatch* spent 493 minutes in total over the four days. During this period, they accessed creature clues 57 times, added 171 documentation records, answered 162 questions correctly out of 204 attempts, and played 3850 mini-game sessions throughout the study. It should be noted that we later realized a few children discovered a glitch in one of the mini-games, which allowed each mini-game session to be completed at an extremely fast rate (approximately 3-6 times per second). This likely contributed to the increased mini-game session count by some children, though this overall does not dominate the time spent on *Snatch and Hatch*.

7.1 Intrinsic Motivation

Children's responses to the interest/enjoyment subscale of the IMI reflected their motivation in using *Snatch and Hatch*. The distribution of scores for questions Q1 to Q6, as shown in Figure 5, suggests that they value enjoyment, fun and interest (with distributions being reversed for negative questions). In prior work, Carbonell et al. [6] investigated motivation using IMI subscales and considered an overall subscale score exceeding the neutral score of 4 as satisfactory. Upon conducting a Mann-Whitney U test to validate the significance of the overall subscale score in our first IMI against a neutral subscale score of 4, we found the interactions with *Snatch and Hatch* to be highly intrinsically motivating and significant ($M = 6.07$, $STD = 1.40$, $z = 8.1078$, $p < 0.01$). We also verified that this motivation was sustained over the four days by running a Wilcoxon signed rank test between the overall interest/enjoyment

¹³<https://www.qualtrics.com/>

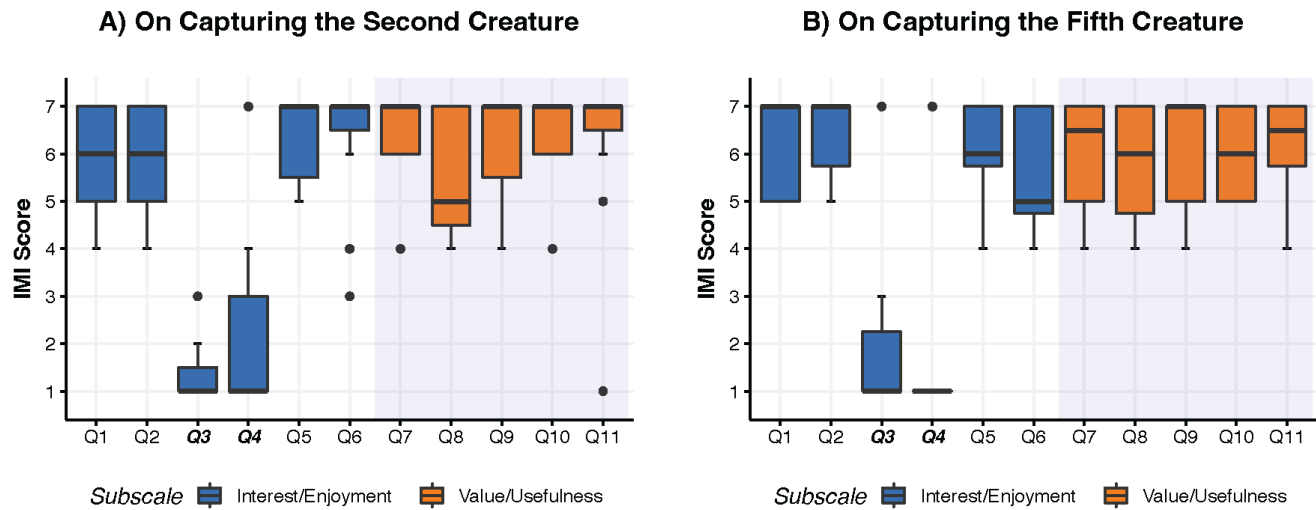


Figure 5: Responses to the Intrinsic Motivation Inventory (IMI) interest/enjoyment (Q1-Q6) and value/usefulness (Q7-Q11) sub-scales A) on capturing the second creature; B) on capturing the fifth creature. Questions which have been emphasised (Q3, Q4) represent negatively-worded questions whose scores must be reversed. See appendix for the complete questionnaire.

scores of the first and second IMIs ($z = -0.1651, p = 0.86$). No significant differences were found, confirming that *Snatch and Hatch* is consistent in its ability to motivate children through its gameplay.

These findings were further reflected in the interviews and observations of children. Most children (7/11) said they enjoyed hatching eggs and finding creatures. P10 said it was “*fun to make different noises*”, although they did not explicitly refer to the process as ‘learning’. Mini-games were the most popular feature among the children. Between creature releases, children were observed playing the mini-games 3889 times, with some participants (P1 and P2) playing the games 1018 and 966 times, respectively, over the four-day study period. In addition, we implemented a feedback form within the game (Figure 2B, top right corner), where children were able to communicate any feedback to us without prior solicitation. The form included a smileyometer[29], which followed a 5 point visual scale ranging from “confusing” to “great”, and an optional text form for children to explain why they responded in this way, as well as anything they would add or change to the game. Children were not required to complete this form. However, we received 9 comments through this such as “*It was interesting and there were lots of games*”, “*It teaches sound*”, “*It is a fun way to learn about sound*”. These are consistent with the IMI scores and follow-up interviews.

7.2 Attitude Towards Science

The value/usefulness subscale of the IMI is commonly regarded as a positive predictor of self-regulatory capabilities in internalisation studies [9]. With questionnaire items Q7 to Q11 (Figure 5) capturing how children view science, we used the subsequent responses to quantify their attitudes on the subject. We adopt the same approach that was used to analyse intrinsic motivation scores to evaluate

attitudes towards science. A Mann-Whitney U test was used to initially check for significance against a neutral subscale score of 4. Our results were significantly above the neutral subscale score, indicating children viewed science as a constructive and beneficial endeavour ($M = 6.10, STD = 1.30, z = 7.5593, p < 0.01$). The average subscale score ($M = 6.14, STD = 1.14$) slightly increased over the four days. However, the Wilcoxon signed rank test indicated no significant differences between the overall subscale scores of the first and second questionnaires ($z = -0.3767, p = 0.70$). We supplemented our findings with our follow-up interviews. When children were asked about the goal of the game, many (6/11) responded that the game was for learning about sound. A few children (2/11) further identified this as learning science, while others (3/11) talked about hatching eggs and collecting creatures. While the majority of children found hatching eggs and finding creatures to be the most fun part of the game, they nevertheless recognised the educational value of the game. We could relate some of our observations to tenets of the Nature of Science as discussed below.

The empirical nature of science suggest that science is based on observations and measurements. One child (P5), while catching a creature requiring a low decibel value, was in a room with their pet dog barking in the background. They quickly realised “*ooh Stella [name of dog] could actually help*”, suggesting that the process of engaging students through creating their own observations may provide opportunities to reflect on related sounds in their daily lives. When children were interviewed during post-study interviews, most expressed that they were able to learn about sound, with one child (P11) saying “*the louder the noises, the higher the decibels. I did not know about decibels before this.*” By making their own observations through the creature discovery feature, children are beginning to form an understanding of sound. This also related to

science being tentative where measurements of sound are not so clear-cut.

We also observed various creative ways of vocalisation to reach the required decibel range. While for some children this was fun because “we’re not usually allowed to scream at home” (P5) or interesting because “I know how loud my brother can scream” (P3), children also found other creative ways to discover and hatch their creatures. Examples from our observations or interviews showed children turning on the bathroom tap, slamming the door, or using the washing machine “so I didn’t have to do anything” (P10). These observations suggest there is an opportunity to use the conditioning provided in *Snatch and Hatch* to make children more receptive towards the Nature of Science by explicitly embodying its themes into the application itself.

7.3 Limitations

In our formative investigations, parents were concerned about their children’s screen time. Although we did not specifically request for additional screen from parents for the study, some parents did provide additional screen time for their child. This may contribute to some of the children’s usage, as one parent said “using the iPad itself is fun for the kids” during formative interviews. As the study was conducted with a small sample size over a period of four days and potentially compounded by sibling influences or other home factors, a further longitudinal study may benefit to minimise these factors.

7.4 Future Work

In the current version of *Snatch and Hatch*, children can only measure the amplitude of sound. However, there are many other built-in sensors that can measure phenomena such as sound frequency, light, magnetic field, or the positioning and orientation of the mobile device. These create further possibilities to expand features of *Snatch and Hatch* and promote playful experiences for children to engage with data they otherwise may not have access to.

8 CONCLUSION

We describe the design and development of *Snatch and Hatch*, a tablet application that engages young children in a playful learning experience, exposing them to the Nature of Science (NoS) through sound. Through a formative investigation and an iterative design process, we developed a production-level application allowing children to use it in their free time at home. We constructed the game play elements to connect with the core tenets of NoS to support children in developing an association between science concepts and the real world. Evaluation of *Snatch and Hatch* in a multi-day study with 11 children (aged 8–12) revealed that our application was motivating and successfully retained this perspective over time, although further testing is still required to evaluate engagement over longer periods. Children reported that science learning is a valuable activity and expressed intent to engage in its concepts in the classroom. Given the enthusiastic response, we see great value in exploring the use of carefully designed playful experiences to create opportunities for implicit learning and improve children’s receptivity towards a formal education involving the principles of the Nature of Science.

9 SELECTION AND PARTICIPATION OF CHILDREN

A total of 27 children were recruited, together with 16 parents. Prior to the study, ethical approval was obtained from the Human Participants Ethics Committee of the National University of Singapore (NUS-IRB-2022-587). Participation consent was received from parents and their children.

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A APPENDIX: IMI QUESTIONNAIRE USED IN THE STUDY

Questionnaire used to measure Intrinsic motivation, adapted from [1]. Each of the statements below were rated by the participants from Not true at all (1) to Very true (7).

Interest/Enjoyment

- Q1. I enjoyed playing *Snatch and Hatch* very much.
- Q2. *Snatch and Hatch* was fun to play.
- Q3. I thought *Snatch and Hatch* was a boring game. (R)
- Q4. *Snatch and Hatch* did not hold my attention at all. (R)

Q5. I would describe *Snatch and Hatch* as very interesting.

Q6. While I was playing *Snatch and Hatch*, I was thinking about how much I enjoyed it.

Value/Usefulness

Q7. I think that playing *Snatch and Hatch* is useful for learning science.

Q8. I think *Snatch and Hatch* is important to do because it can help me with learning science.

Q9. I think playing *Snatch and Hatch* could help me to learn science.

Q10. I believe learning about sound could be beneficial to me.

Q11. I think learning about sound is an important activity.